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# UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

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Anonymous submission

## ABSTRACT

The computational complexity of deriving formal explanations for the predictions of machine-learning models has been investigated principally for two fundamental explanation types: (i) *sufficient reasons*, subsets of input features that, once fixed, determine a prediction irrespective of the remaining features, and (ii) *contrastive reasons*, subsets of input features whose modification suffices to overturn a prediction. These explanations have hitherto been studied in isolation, across disparate settings that separately oppose non-probabilistic to probabilistic formulations and local to global reasoning. We introduce a unifying framework that subsumes all of these settings simultaneously, establishing that each explanation task is precisely characterized as the minimization of a single probabilistic value function. Building on this characterization, we show that the complexity of the resulting computation is governed by three canonical properties of that value function — (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — three of the structural properties central to *combinatorial optimization*. Our analysis exposes a sharp and counter-intuitive dichotomy in how these properties manifest across explanation settings: whereas the *local* value functions possess none of them, we prove that their *global* counterparts acquire monotonicity together with the appropriate modularity structure. We exploit this dichotomy to derive a series of polynomial-time results for computing global explanations with provable guarantees — under standard distributional assumptions, notably empirical and independent distributions — uniformly across model families that span the interpretability spectrum, from neural networks and tree ensembles to decision trees. In sharp contrast, we prove that even drastically simplified instances of these same explanations remain NP-hard in the corresponding local setting.

## 1 INTRODUCTION

Despite substantial progress in methods for explaining the decisions of ML models, a recurring and by now well-documented finding is that many of the most desirable explanation types — those equipped with provable guarantees — are computationally hard to obtain (Barceló et al., 2020; Van den Broeck et al., 2022), and this difficulty intensifies precisely in the complex, highly non-linear models for which reliable explanations are most consequential (Barceló et al., 2020; Adolfi et al., 2025; 2024). Consequently, the *computational complexity* of obtaining explanations has emerged as a central theoretical concern, and a rapidly expanding body of work now seeks to delineate precisely which explanation types can be computed efficiently for which model classes, and which remain provably out of reach (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

**From sufficient to contrastive reasons.** Within this literature, two explanation types stand out as the most extensively studied objects of complexity-theoretic analysis: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features  $S$  whose fixed values alone determine the prediction: once the features in  $S$  are held at their given values, the output is invariant to every assignment of the complementary set  $\bar{S}$ . A *contrastive reason* is, dually, a subset of input features  $S$  whose modification suffices to overturn the prediction.

Unlike additive attribution methods, which distribute continuous importance scores across features yet are frequently difficult for humans to interpret (Kumar et al., 2020) and offer limited actionability (Bilodeau et al., 2024), sufficient and contrastive reasons furnish *discrete, condition-like explanations* that answer, directly and unambiguously, “what is enough to justify this prediction?” and “what must change to overturn it?”. This directness underlies their central position among classical XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), their demonstrated superiority over additive models in improving human prediction (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and their established utility for downstream tasks such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A further well-established principle holds that *smaller* sufficient and contrastive explanations are more interpretable, which elevates *minimality* to a central guarantee of interest in its own right (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021). Together, these considerations position minimal sufficient and contrastive reasons as a natural focal point for a complexity-theoretic account of formal explanations.

One of the earliest complexity-theoretic studies of sufficient and contrastive reasons is due to Barceló et al. (2020), who established that computing a minimum-size sufficient reason is NP-hard for decision trees and, more severely,  $\Sigma_2^P$ -hard for neural networks; for contrastive reasons, the smallest explanation is computable in polynomial time for decision trees but becomes NP-hard for neural networks. Analogous hardness results were subsequently extended to tree ensembles (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b), confirming that these obstructions persist across the standard model families and are not artifacts of any single representation.

**From non-probabilistic to probabilistic explanations.** A recurring criticism of the classical definition of sufficient and contrastive reasons targets its rigidity: because the conditions must hold in an absolute sense over the entire input domain, the resulting explanations are frequently excessively large or altogether uninformative (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To relax this rigidity, the literature has converged on a strictly more general, *probabilistic* formulation of sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023), in which one seeks subsets of input features that determine or overturn a prediction with probability exceeding a prescribed threshold  $\delta$ ; the non-probabilistic notions are recovered exactly at  $\delta = 1$ .

Wäldchen et al. (2021) initiated the complexity-theoretic study of probabilistic sufficient reasons, proving the problem  $\text{NP}^{\text{PP}}$ -hard for CNF classifiers — a hardness that carries over to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). The decisive theoretical insight of the probabilistic setting is that the underlying probability function *loses monotonicity* (Arenas et al., 2022; Izza et al., 2023), and it is precisely this loss that renders even *subset-minimal* explanations computationally hard. Indeed, Arenas et al. (2022) establish the striking result that computing a subset-minimal probabilistic sufficient reason is NP-hard already for decision trees — in stark contrast to the non-probabilistic case, where the same explanations are computable in polynomial time (Huang et al., 2021; Barceló et al., 2020). This monotonicity loss is the pivot on which our own results turn.

**From local to global explanations.** More recently, Bassan et al. (2024) broadened the complexity analysis from the *local* (non-probabilistic) setting — in which an explanation is tied to a single prediction — to the *global* (non-probabilistic) setting, in which sufficient or contrastive reasons must account for the model’s behavior across the entire input domain. As with other non-probabilistic formulations (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), however, the underlying criteria are exceedingly stringent — and arguably more so globally than locally. In particular, any feature excluded from a global sufficient reason is deemed strictly redundant (Bassan et al., 2024), so that the explanation tends to encompass nearly all input features and thereby forfeits much of its informativeness. This tension — global reasoning offering conceptual leverage yet the non-probabilistic criterion sacrificing informativeness — is precisely what the probabilistic generalization we study is designed to resolve.

## OUR CONTRIBUTIONS

1. **A unifying framework.** We unify the previously fragmented landscape of explanation-computation problems — sufficient and contrastive, probabilistic and non-probabilistic,

local and global — within a single framework, recasting every one of these tasks as the minimization of a common value function subject to a threshold. We then isolate three canonical properties of this value function that decisively govern the complexity of the task: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*. This reformulation subsumes, as special cases, notions previously analyzed in isolation, and it is the vehicle through which the structural results below transfer uniformly across settings.

2. **A local–global structural dichotomy.** We prove that these properties behave in *fundamentally different ways* depending on whether the value function is local or global. Specifically, we establish the counterintuitive fact that, although the *local* value functions for both sufficient and contrastive reasons are *non-monotonic*, their *global* counterparts are *monotonic non-decreasing* — and this holds for *every* underlying distribution, requiring no independence assumption. We further identify modularity structure exclusive to the global setting: under feature independence, the global sufficient value function is *supermodular* and the global contrastive value function is *submodular*, whereas the corresponding local value functions possess neither property even under a uniform distribution. Averaging over inputs, in other words, is what manufactures the combinatorial structure that local reasoning lacks.
3. **Tractability of global explanations, hardness of local ones.** We convert this structural dichotomy into complexity results, demonstrating that global explanations with provable guarantees can be computed efficiently precisely where their local counterparts are provably intractable. We establish these results uniformly across three widely used model families that span the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. We first prove that, while computing a subset-minimal local probabilistic sufficient/contrastive explanation is NP-hard already for decision trees (Arenas et al., 2022), its global counterpart is computable in polynomial time whenever the value function admits polynomial-time evaluation — as it does, for instance, under independent distributions. We then extend this tractability to *any* model with polynomial-time inference — neural networks and tree ensembles included — under empirical distributions: computing a subset-minimal global sufficient/contrastive explanation is polynomial-time solvable, whereas the local version remains NP-hard for these very models.
4. **Constant-factor approximation, and matching local inapproximability.** Finally, we prove a still stronger separation by exploiting the submodularity and supermodularity of the global value functions — structure that is entirely absent locally. We show that, for a fixed empirical distribution, cardinally minimal global explanations admit provable *constant-factor* approximation guarantees even for complex models such as neural networks and tree ensembles. In sharp contrast, we establish strong inapproximability for the local setting, proving that *no bounded* approximation is attainable even under drastically simplified assumptions — indeed already for a single data point.

We regard the unifying value-function formulation, and the local–global structural dichotomy it exposes, as the paper’s primary conceptual contributions; the model-specific tractability and inapproximability results are the concrete separations that this structure makes possible. Owing to space constraints, the main text states our principal theorems and corollaries and conveys the ideas behind them, deferring the complete proofs to the appendix.

## 2 PRELIMINARIES

**Setting.** We consider an input space of dimension  $n \in \mathbb{N}$ , with input vectors  $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ , each coordinate  $\mathbf{x}_i$  ranging over its own domain  $\mathcal{X}_i$ , discrete or continuous. The input feature space is accordingly the product  $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$ , and we study classification models  $f : \mathbb{F} \rightarrow [c]$ , where  $c \in \mathbb{N}$  denotes the number of classes. We further fix a *generic* distribution  $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$  over the input space. Since accurately approximating  $\mathcal{D}$  is computationally infeasible in many settings, a standard and practical alternative substitutes a fixed empirical dataset  $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$  as a proxy for the “true” distribution — obtained, for instance, by sampling a subset of the available training data.

The explanations we study come in local and global forms. A local explanation targets a specific prediction  $\mathbf{x} \in \mathbb{F}$ , articulating why the model assigned the label  $f(\mathbf{x})$  to that particular instance. A global explanation, by contrast, abstracts away from any single  $\mathbf{x}$ : it characterizes the reasoning

that governs the behavior of  $f$  across the input space, thereby capturing the model’s decision logic as a whole. This local–global axis, orthogonal to the sufficient–contrastive and probabilistic–non-probabilistic distinctions, is the one along which our central dichotomy emerges.

**Models.** Although many of our results are model-agnostic — the intrinsic properties of the value functions being a prime example — we also derive model-specific complexity results for widely used ML models. To span the full interpretability spectrum, we deliberately analyze models ranging from those regarded as “black-box” to those regarded as “interpretable”: (i) decision trees, (ii) neural networks (encompassing every architecture at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, covering both majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). The formal definitions are deferred to the Appendix.

**Distributions.** In probabilistic explanation settings the complexity depends critically on the input distribution  $\mathcal{D}$ , and we accordingly organize our analysis around three distribution classes, ordered from most to least general: (i) *general distributions*, which impose no assumption on  $\mathcal{D}$  whatsoever and hence subsume all of the following — the setting in which we establish the properties that hold universally, most notably monotonicity; (ii) *empirical distributions*, comprising all distributions that place uniform mass on the finite dataset  $\mathbf{D}$ , a modeling choice ubiquitous in XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, which posit that the features of  $\mathcal{D}$  are mutually independent — likewise a widely adopted assumption in the XAI literature (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). We emphasize that these last two classes are distinct rather than nested: an empirical distribution places all its mass on the observed points of  $\mathbf{D}$  and therefore, in general, encodes rich inter-feature dependencies and is *not* product-distributed, whereas an independent distribution factorizes across features by construction (Van den Broeck et al., 2022). Our supermodularity, submodularity, and constant-factor approximation results invoke feature independence explicitly, and we are careful throughout to state which distributional assumption each result requires. The complete formal definitions appear in the Appendix.

### 3 FORMS OF REASONS

This section introduces the explanation types studied throughout the paper. We begin with the strict non-probabilistic definitions of (local and global) sufficient and contrastive reasons, and then generalize them to their more flexible probabilistic counterparts, which recover the non-probabilistic notions exactly as the special case  $\delta = 1$ . Isolating these four axes — sufficient versus contrastive, local versus global — at the level of definitions is what later permits their common treatment as a single optimization template.

#### 3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

**Sufficient reasons.** In *feature selection*, one commonly identifies the top  $k$  features responsible for a model’s decision. We adopt the well-established *sufficiency* criterion for this selection, which underlies many widely used explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022) and applies equally at the local level — for a single prediction — and at the global level — across the entire input domain. Following standard conventions, a *local sufficient reason* is a subset  $S \subseteq \{1, \dots, n\}$  such that fixing the features in  $S$  to their values in  $\mathbf{x}$  preserves the prediction  $f(\mathbf{x})$  under every assignment of the complementary features  $\bar{S}$ . Formally,  $S$  is a local sufficient reason for  $\langle f, \mathbf{x} \rangle$  iff:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here  $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$  denotes the vector whose features in  $S$  take their values from  $\mathbf{x}$  and whose features in  $\bar{S}$  take their values from  $\mathbf{z}$ . Local sufficient reasons are intimately connected to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which seek an alternative input  $\mathbf{z}'$  that preserves the prediction (i.e.,  $f(\mathbf{z}') = f(\mathbf{x})$ ) while lying as “close” as possible to the original point: instantiating  $\mathbf{z}'$  as  $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$  and measuring proximity by Hamming distance makes the two notions coincide exactly, so that our results speak directly to that literature as well.

A *global sufficient reason* (Bassan et al., 2024) is a subset  $S \subseteq \{1, \dots, n\}$  that is simultaneously a local sufficient reason for *every* input  $\mathbf{x} \in \mathbb{F}$ :

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

**Contrastive reasons.** A second, equally prevalent explanatory paradigm identifies subsets of input features whose alteration *overturns* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024), thereby pinpointing the minimal intervention required to change the model’s output. Formally, a subset  $S \subseteq \{1, \dots, n\}$  is a *local contrastive reason* for  $\langle f, \mathbf{x} \rangle$  iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons are correspondingly tied to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment  $\mathbf{x}'$  at which the prediction flips, i.e.,  $f(\mathbf{x}') \neq f(\mathbf{x})$ ; under the Hamming-weight distance metric, the two notions again coincide exactly.

As with sufficient reasons, the contrastive notion admits a global analogue: a subset of input features whose alteration can overturn the prediction at every input across the domain of interest. This global form is closely aligned with methods for identifying bias (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and with *group* counterfactual explanation methods, which seek counterfactuals spanning multiple data instances (Carrizosa et al., 2024; Warren et al., 2024), so that the global contrastive setting is of independent practical interest. Formally,  $S$  is a *global contrastive reason* with respect to  $f$  iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

### 3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As noted in the introduction, non-probabilistic sufficient and contrastive reasons are widely criticized for imposing conditions that are far too stringent, motivating the shift to *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021). These definitions relax the requirement so that the relevant guarantee need hold only with probability at least  $\delta \in [0, 1]$ ; they strictly generalize the original notions, which are recovered exactly at  $\delta = 1$ , and it is this parameterization by  $\delta$  that our framework exploits.

**Probabilistic sufficient reasons.** We call  $S \subseteq \{1, \dots, n\}$  a *local  $\delta$ -sufficient reason* for  $\langle f, \mathbf{x} \rangle$  if fixing the features in  $S$  to their values in  $\mathbf{x}$  and sampling the complementary features from  $\mathcal{D}$  preserves the classification with probability at least  $\delta$ ; that is:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where  $\mathbf{z}_S = \mathbf{x}_S$  indicates that the features of  $\mathbf{z}$  in  $S$  are fixed to their values in  $\mathbf{x}$ . Following the standard construction of global explanations — averaging over all inputs in the domain — we call  $S \subseteq \{1, \dots, n\}$  a *global  $\delta$ -sufficient reason* for  $\langle f \rangle$  if the expectation of the local sufficiency probability, taken over samples  $\mathbf{x}$  drawn from  $\mathcal{D}$ , is itself at least  $\delta$ ; that is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

**Probabilistic contrastive reasons.** Mirroring the non-probabilistic case, a *local  $\delta$ -contrastive reason* is a subset of input features whose variation overturns the prediction with sufficiently high probability. Dually to sufficient reasons, here it is the *complementary* set  $\bar{S}$  that is fixed to its values in  $\mathbf{x}$ , while the features in  $S$  are permitted to vary, and we require that the prediction differ from  $f(\mathbf{x})$  with probability at least  $\delta$ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

In the global setting, and analogously to global sufficient reasons,  $S$  is a *global  $\delta$ -contrastive reason* when the expectation of the local contrastive probability, taken over samples  $\mathbf{x}$  drawn from  $\mathcal{D}$ , is at least  $\delta$ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

## 4 FORMS OF MINIMALITY

As discussed in the introduction, all of the explanation forms considered so far — sufficient or contrastive, local or global — share the premise, standard in the literature, that *smaller* explanations are more meaningful, which elevates their *minimality* to a provable guarantee of primary interest. We study two central notions of minimality, both of which apply uniformly to every explanation type in our framework:

**Definition 1.** Assuming a subset  $S \subseteq \{1, \dots, n\}$  is an explanation, then:

1.  $S$  is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff  $S$  has the *smallest explanation cardinality*  $|S|$  (i.e., there is no explanation  $S'$  such that  $|S'| < |S|$ ).
2.  $S \subseteq \{1, \dots, n\}$  is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff  $S$  is an explanation, and any  $S' \subsetneq S$  is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal  $S$  is subset-minimal, but the converse fails (Appendix B.3 supplies an explicit example). We adopt the terms subset- and cardinally minimal, rather than local and global minima, precisely to forestall conflation with local versus global explanations, which concern input-level versus domain-level reasoning. Both notions apply to *every* explanation form we analyze; a cardinally minimal local probabilistic  $\delta$ -sufficient reason, for instance, is one of smallest cardinality  $|S|$ , whereas a subset-minimal one is any qualifying  $S$  admitting no qualifying proper subset  $S' \subsetneq S$ . The two notions carry markedly different computational profiles, and our results treat each explicitly.

## 5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

The central observation of this work is that all of the computational problems introduced above — local or global, sufficient or contrastive, probabilistic or non-probabilistic — collapse into a single template: find a minimum-size subset  $S$  satisfying  $v(S) \geq \delta$ , with  $\delta := 1$  recovering the non-probabilistic settings. To make this precise, we fix notation for the four value functions. Let  $v_{\text{suff}}^\ell$  denote the local sufficiency probability of equation 5, namely  $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ , and let  $v_{\text{suff}}^g$  denote its global variant of equation 6; analogously, let  $v_{\text{con}}^\ell$  and  $v_{\text{con}}^g$  denote the local and global contrastive probabilities of equations 7 and 73, respectively. With this notation in hand, the unified task of computing a cardinally minimal local/global sufficient/contrastive  $\delta$ -reason is stated as follows:

### Cardinally Minimal $\delta$ -Explanation:

**Input:** Model  $f$ , a distribution  $\mathcal{D}$ , (possibly, an input  $\mathbf{x}$ ), a *general* value function  $v : 2^n \rightarrow [0, 1]$  (defined using  $f$ ,  $\mathcal{D}$ , and possibly  $\mathbf{x}$ ), and some  $\delta \in [0, 1]$ .

**Output:** A subset  $S \subseteq [n]$  such that  $v(S) \geq \delta$  and  $|S|$  is minimal.

The relaxed variant, in which the objective is a subset-minimal rather than a cardinally minimal local or global sufficient/contrastive reason, is captured by the following optimization task:

### Subset Minimal $\delta$ -Explanation:

**Input:** Model  $f$ , a distribution  $\mathcal{D}$ , (possibly, an input  $\mathbf{x}$ ), a *general* value function  $v : 2^n \rightarrow [0, 1]$  (defined using  $f$ ,  $\mathcal{D}$ , and possibly  $\mathbf{x}$ ), and some  $\delta \in [0, 1]$ .

**Output:** A subset  $S \subseteq [n]$  such that  $v(S) \geq \delta$  and for any  $S' \subsetneq S$  it holds that  $v(S') < \delta$ .

**Properties of  $v$  that govern the complexity.** We now isolate the three properties of the value function that, as we later prove, determine the complexity of computing explanations. The first is *non-decreasing monotonicity*, which guarantees that every marginal contribution  $v(S \cup \{i\}) - v(S)$  is non-negative. Formally:

**Definition 2.** We say that a value function  $v$  maintains non-decreasing monotonicity if for any  $S \subseteq \{1, \dots, n\}$  and any  $i \in \{1, \dots, n\}$  it holds that:  $v(S \cup \{i\}) \geq v(S)$ .

The remaining two properties, *supermodularity* and *submodularity*, characterize how the marginal contribution  $v(S \cup \{i\}) - v(S)$  evolves as the conditioning set grows: it is non-decreasing in the supermodular case and, dually, non-increasing in the submodular case. These are exactly the diminishing-

and increasing-returns conditions on which the approximation guarantees of combinatorial optimization rest. Formally:

**Definition 3.** *Let there be some value function  $v$ , some  $S \subseteq S' \subseteq \{1, \dots, n\}$ , and  $i \notin S'$ . Then:*

1.  *$v$  maintains supermodularity iff it holds that:  $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$ .*
2.  *$v$  maintains submodularity iff it holds that:  $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$ .*

## 6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

Prior work shows that in the local, non-probabilistic setting, subset-minimal sufficient or contrastive reasons are computable precisely because the value function is monotone — but this monotonicity holds only in the restricted regime  $\delta = 1$ , whose rigidity sharply curtails its practical value. One might hope to recover flexibility by passing to probabilistic guarantees for arbitrary  $\delta$ ; prior results show, however, that monotonicity collapses in this regime, and it is exactly this collapse that renders computing explanations harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b). The results of this section establish that the collapse is confined to the *local* setting: globally, monotonicity survives for every  $\delta$ .

In the still more extreme *global, non-probabilistic* setting, Bassan et al. (2024) established a stringent *uniqueness* property: there exists exactly *one* subset-minimal explanation. Yet the requirement  $\delta = 1$  renders this setting highly restrictive — especially globally, where the sufficiency condition must hold for *all* inputs — and indeed Bassan et al. (2024) show that the unique minimal subset coincides with the set of all non-strictly-redundant features, which in practice may be all of them. We prove that this uniqueness is an artifact of the  $\delta = 1$  regime: in the general global *probabilistic* setting, for arbitrary  $\delta$ , uniqueness *fails*, and the number of subset-minimal explanations can be as large as *exponential*. This failure of uniqueness is not a defect but a feature: it is precisely the abundance of admissible explanations at  $\delta < 1$  that leaves room for genuine minimization.

**Proposition 1.** *While the non-probabilistic case ( $\delta = 1$ ) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary  $\delta$ ), there exist functions  $f$  that have  $\Theta(\frac{2^n}{\sqrt{n}})$  subset-minimal global sufficient reasons.*

Strikingly, although uniqueness is confined to the special case  $\delta = 1$ , the crucial *monotonicity* property is not: we prove that it holds for the global value function across *all* values of  $\delta$ , for both the sufficient ( $v_{\text{suff}}^g$ ) and contrastive ( $v_{\text{con}}^g$ ) settings, and — as our proof makes explicit — under *any* underlying distribution. This is a clean reversal of the local picture, in which the corresponding value functions  $v_{\text{suff}}^l$  and  $v_{\text{con}}^l$  are non-monotonic:

**Proposition 2.** *While the local probabilistic setting (for any  $\delta$ ) lacks monotonicity — i.e., the value functions  $v_{\text{con}}^l$  and  $v_{\text{suff}}^l$  are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any  $\delta$ ), both value functions  $v_{\text{con}}^g$  and  $v_{\text{suff}}^g$  are monotonic non-decreasing.*

Beyond the fact that monotonicity holds globally but not locally, the global setting possesses additional structure of its own — namely submodularity and supermodularity. Concretely, under the standard assumption of feature independence, the global sufficient value function  $v_{\text{suff}}^g$  is supermodular, whereas its local counterpart  $v_{\text{suff}}^l$  fails to be supermodular even under the far more restrictive assumption of a uniform (and hence independent) input distribution. The separation is therefore not attributable to the distribution: it is intrinsic to the local–global axis.

**Proposition 3.** *While the local probabilistic sufficient setting (for any  $\delta$ ) lacks supermodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{suff}}^l$  is not supermodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{suff}}^g$  is supermodular.*

For the complementary family of settings — global probabilistic *contrastive* reasons — the value function exhibits, under the same independence assumption, the dual structure: not supermodular but *submodular*. This is again in pointed contrast to the local setting, where the corresponding value function is neither submodular nor supermodular, and it is this submodularity that later yields the sharper of our two approximation guarantees.

**Proposition 4.** *While the local probabilistic contrastive setting (for any  $\delta$ ) lacks submodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{con}}^{\ell}$  is not submodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{con}}^g$  is submodular.*

## 7 COMPUTATIONAL COMPLEXITY RESULTS

### 7.1 SUBSET MINIMAL EXPLANATIONS

We now analyze the complexity of computing *subset-minimal* explanations — local or global, sufficient or contrastive — across the model families under study. The decisive property here is the *monotonicity* of the value function: the preceding section established that both global value functions  $v_{\text{con}}^g$  and  $v_{\text{suff}}^g$  are monotone, whereas the local functions  $v_{\text{con}}^{\ell}$  and  $v_{\text{suff}}^{\ell}$  are not. Precisely this gap makes a greedy procedure converge to a subset-minimal explanation globally while failing to do so locally, and it yields our first separation: computing subset-minimal global explanations is polynomial-time solvable, while computing their local forms is hard. We begin with the following generalized greedy procedure:

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**Algorithm 1** Subset Minimal Explanation Search

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**Input** Value function  $v$ , and some  $\delta \in [0, 1]$

- 1:  $S \leftarrow \{1, \dots, n\}$
- 2: **while**  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  **do**
- 3:    $j \leftarrow \arg\max_{i \in S} v(S \setminus \{i\})$
- 4:    $S \leftarrow S \setminus \{j\}$
- 5: **end while**
- 6: **return**  $S$   $\triangleright S$  is a (subset minimal?)  $\delta$ -explanation

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Algorithm 1 computes a subset-minimal  $\delta$ -explanation by top-down greedy elimination. It initializes  $S$  to the full feature set  $\{1, \dots, n\}$  and, at each iteration, tests whether the smallest attainable value  $\min_{i \in S} v(S \setminus \{i\})$  still meets the threshold  $\delta$ ; if so, it removes the feature  $j$  whose deletion decreases the value function the least — equivalently, the feature  $j$  maximizing  $v(S \setminus \{j\})$  — and repeats. The procedure returns  $S$  once no further feature can be removed without violating the threshold.

Whether Algorithm 1 in fact returns a subset-minimal explanation hinges entirely on the *monotonicity* of  $v$ . At termination, removing any single feature  $i$  from  $S$  drives  $v(S \setminus \{i\})$  below  $\delta$ ; monotonicity strengthens this per-feature condition into a global one, guaranteeing that  $v(S \setminus S') < \delta$  for *every* nonempty  $S' \subseteq S$  — precisely the certificate of subset minimality. Invoking the monotonicity results of the previous section, we obtain the following:

**Proposition 5.** *Computing Algorithm 1 with the local value functions  $v_{\text{con}}^{\ell}$  and  $v_{\text{suff}}^{\ell}$  does not always converge to a subset minimal  $\delta$ -sufficient/contrastive reason. However, computing it with the global value functions  $v_{\text{con}}^g$  or  $v_{\text{suff}}^g$  necessarily produces subset minimal  $\delta$ -sufficient/contrastive reasons.*

Building on this convergence guarantee, we now establish complexity results for computing the various forms of subset-minimal explanations within our framework, treating each of the analyzed model families in turn.

**Decision trees.** We begin with a deliberately simple, ostensibly “interpretable” scenario in which  $f$  is a decision tree and  $\mathcal{D}$  is an independent distribution — and even here a strict local–global separation emerges. Arenas et al. (2022) proved the surprising intractability result that, unless  $\text{PTIME}=\text{NP}$ , no polynomial-time algorithm computes a subset-minimal local  $\delta$ -sufficient reason for decision trees under independent distributions (indeed, already under the uniform distribution). We prove that the very same problem becomes tractable globally: a subset-minimal global  $\delta$ -sufficient reason for decision trees is computable in polynomial time. Formally:

**Theorem 1.** *If  $f$  is a decision tree and the probability term  $v_{\text{suff}}^g$  can be computed in polynomial time given the distribution  $\mathcal{D}$  (which holds for independent distributions, among others), then obtaining a subset-minimal global  $\delta$ -sufficient reason can be obtained in polynomial time. However, unless  $\text{PTIME}=\text{NP}$ , no polynomial-time algorithm exists for computing a local  $\delta$ -sufficient reason for decision trees even under independent distributions.*

**Extension to other tractable models.** The same tractability guarantee for global explanations extends to *orthogonal DNFs* (Crama & Hammer, 2011) — disjunctive normal form formulas whose terms are pairwise mutually exclusive. The extension is consequential: orthogonal DNFs strictly generalize decision trees while preserving the structural properties on which the tractability argument depends, so the guarantee reaches a strictly broader class than tree-structured models alone. The full argument appears in Appendix N.

**Neural networks, tree ensembles, and other complex models.** We now carry these results well beyond decision trees. We prove that whenever  $\mathcal{D}$  is an empirical distribution and  $f$  admits polynomial-time inference — a mild assumption satisfied by essentially every model deployed in practice — computing a subset-minimal global  $\delta$ -sufficient or  $\delta$ -contrastive reason is polynomial-time solvable, irrespective of the internal complexity of the model. Here the empirical distribution does the work that independence did for decision trees: it renders the value function exactly and efficiently evaluable.

**Proposition 6.** *For any model  $f$ , and empirical distribution  $\mathcal{D}$  — computing a subset-minimal global  $\delta$ -sufficient or  $\delta$ -contrastive reason for  $f$  can be done in polynomial time.*

This guarantee — valid for *any* model with polynomial-time inference under an empirical distribution — sets up a further sharpening of the local–global divide. For sufficiently expressive models, computing subset-minimal *local* explanations remains intractable even under empirical distributions; we establish this for both neural networks and tree ensembles, obtaining the following strict complexity separation:

**Theorem 2.** *Assuming  $f$  is a neural network or a tree ensemble, and  $\mathcal{D}$  is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global  $\delta$ -sufficient and contrastive reasons. However, unless  $\text{PTIME}=\text{NP}$ , there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -sufficient reason or a subset minimal local  $\delta$ -contrastive reason.*

## 7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now turn from subset-minimal reasons to the considerably harder task of computing a *cardinally minimal*  $\delta$ -sufficient/contrastive reason. In the global setting, the interplay of monotonicity with sub-/supermodularity yields *provable approximation* guarantees for these explanations, whereas the local setting remains not merely intractable but inapproximable. This deepens the local–global divide established above, extending it from exact computation to approximation.

**A unified greedy approximation algorithm.** For a monotone non-decreasing submodular value function, computing a cardinally minimal explanation is precisely an instance of the *submodular set cover* problem (Wolsey, 1982); for a *supermodular* function, it becomes the non-submodular variant of that problem (Shi et al., 2021). Both problems are classical and are valued for the strong approximation guarantees they admit (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). The standard tool for submodular set cover is a classic greedy algorithm, which we present as Algorithm 2; it serves as the common foundation for approximating cardinally minimal sufficient and contrastive  $\delta$ -reasons alike, and we analyze its guarantees for each case in turn below.

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### Algorithm 2 Cardinally Minimal Explanation Approximation Search

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**Input** Value function  $v$ , and some  $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$  ▷  $|S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

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Algorithm 2 is the bottom-up counterpart of Algorithm 1: rather than eliminating features from the full set, it grows an initially empty subset, at each step appending the feature that maximizes the increase in  $v(S \cup i)$ , and halts as soon as the accumulated value first meets the threshold  $\delta$ .

**Cardinally minimal contrastive reasons.** Consider global contrastive reasons first. Since both monotonicity and submodularity hold here (the latter under feature independence), the task reduces exactly to *submodular set cover*, and Algorithm 2 inherits the classic guarantee of Wolsey (1982): a

Harmonic-based factor for integer-valued objectives and, more generally, an  $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution  $\mathcal{D}$  of fixed sample size, this ratio is at most logarithmic in the sample size  $|D|$  and hence *constant* in the input dimension — a genuine, if worst-case, guarantee. The *local* setting behaves in diametric opposition: even for a *single* sample point, no *bounded* approximation exists. The contrast between a constant factor and no bounded factor is, we stress, the sharpest form the local–global divide can take.

**Theorem 3.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by  $\mathcal{O}(\ln(|D|))$ , for computing a global cardinally minimal  $\delta$ -contrastive reason for  $f$ , assuming feature independence. In contrast, unless  $\text{PTIME}=\text{NP}$ , no bounded approximation exists for computing a local cardinally minimal  $\delta$ -contrastive reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

We further establish a matching *lower bound* for Theorem 3 in the case of an empirical distribution with no independence assumption, confirming that the  $\mathcal{O}(\ln(|D|))$  guarantee is essentially tight and not an artifact of a loose analysis (see Appendix M).

**Cardinally minimal sufficient reasons.** Global *sufficient* reasons correspond to the supermodular variant of set cover, which is inherently harder to approximate than the submodular version governing contrastive reasons. It nonetheless admits approximation guarantees whenever the function has bounded curvature (Shi et al., 2021). Recall that the *total curvature* of  $v : 2^n \rightarrow \mathbb{R}$  is  $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$ . Applying the results of Shi et al. (2021), we show that Algorithm 2 attains an  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation — the curvature term  $\frac{1}{1-k^f}$  being the price paid for supermodularity over the submodular case.

Under a fixed empirical distribution this approximation is again constant in the input dimension. Contrastive reasons enjoy the tighter  $\mathcal{O}(\ln(|D|))$  bound, whereas sufficient reasons additionally pay the  $\frac{1}{1-k^f}$  curvature factor; even so, the resulting guarantee remains constant-factor. The local variant, by contrast, stays inapproximable, admitting no *bounded* approximation even when  $|D| = 1$  — so the divide persists in its strongest form for sufficient reasons as well.

**Theorem 4.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal  $\delta$ -sufficient reason for  $f$ , assuming feature independence. In contrast, unless  $\text{PTIME}=\text{NP}$ , there is no bounded approximation for computing a local cardinally minimal  $\delta$ -sufficient reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

Taken together, these results reinforce and extend Subsection 7.1, which established the tractability of subset-minimal global explanations against the intractability of their local counterparts. Here we lift the same dichotomy to the harder cardinally minimal regime: approximating cardinally minimal global explanations is tractable to within a constant factor, whereas their local counterparts admit no bounded approximation at all. The local–global divide therefore holds at both levels of minimality and at both the exact and approximate computational scales.

## 8 LIMITATIONS AND FUTURE WORK

Our central findings — above all those concerning the intrinsic properties of the value functions — hold in full generality, and we instantiate them to obtain concrete complexity results for specific model classes (e.g., neural networks), distributional assumptions (e.g., empirical and independent distributions), and explanation definitions within our framework. We are deliberately explicit about the scope of each result. Two assumptions do substantive work and warrant emphasis. First, the supermodularity, submodularity, and constant-factor approximation results assume *feature independence*; monotonicity, and hence the tractability of subset-minimal global explanations, requires no such assumption and holds for arbitrary distributions. Second, the tractability results presuppose that the value function can be evaluated *exactly* in polynomial time — which holds structurally for decision trees and orthogonal DNFs under independent distributions, and by direct enumeration for any polynomial-time-inference model under empirical distributions — but need not hold for arbitrary distributions. These are the boundaries of the present analysis rather than obstacles to it:

the foundational structural properties and the tractability and intractability separations we establish supply a firm basis from which a substantially broader range of explainability scenarios can be investigated. We view the framework itself as the durable contribution, and the following questions as the natural next steps it opens.

Concretely, our study raises several theoretical questions of independent interest:

1. First, whether the tractability results we obtain for global explainability can be sharpened further for structurally simple model classes — decision trees, linear models, or other inherently tractable models. Although local explanations for such models are frequently intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the global explanation forms studied here carry strong structural properties that already yield markedly improved guarantees and may permit tighter constants still. Whether the worst-case approximation factors under empirical distributions can be improved — or shown tight beyond the contrastive case treated in Appendix M — is a closely related direction.
2. Second, whether the feature-independence hypothesis underlying the modularity results can be weakened to *bounded*, Markov, or conditional-independence structure while retaining submodularity and supermodularity; identifying the exact dependence structure these properties require would delineate their reach precisely.
3. Third, when the relevant expectations cannot be computed *exactly*, whether they can instead be approximated via alternative techniques — Fully Polynomial Randomized Approximation Schemes (FPRAS) for model classes such as DNF formulas (Meel et al., 2019) (and hence for tree ensembles), or Monte Carlo estimation of the kind used in (Subercaseaux et al., 2025). The central challenge is to characterize which tolerance levels preserve the minimality guarantees under approximation error.
4. Finally, whether alternative notions of feature importance — for instance, loss measures such as KL divergence (Conmy et al., 2023) — exhibit the same monotonicity, submodularity, and supermodularity phenomena in the transition from local to global explanations, thereby enabling efficient feature selection under these metrics as well.

## 9 CONCLUSION

We have introduced a unified framework for analyzing diverse explanations and used it to expose a stark divide between local and global sufficient and contrastive reasons. Whereas the *local* variants possess none of the structural properties central to combinatorial optimization — monotonicity, submodularity, or supermodularity — we prove that their *global* counterparts acquire exactly these properties: (i) *monotonicity*, for every distribution; (ii) *submodularity* for contrastive reasons; and (iii) *supermodularity* for sufficient reasons, the latter two under feature independence. These structural results underpin a series of complexity theorems establishing that global explanations with provable guarantees are computable efficiently — exactly for subset-minimal reasons, and to a constant factor for cardinally minimal ones — even for complex model classes such as neural networks and tree ensembles, while their local counterparts remain NP-hard, or outright inapproximable, even in drastically simplified scenarios. The recurring lesson is that averaging over inputs is precisely what confers the combinatorial structure on which efficient computation depends. In charting both the tractable and the intractable frontiers of formal explanation from a single vantage point, our framework supplies a foundation on which subsequent work — on weaker distributional assumptions, approximate value computation, and alternative importance measures — can build.

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# Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

**Appendix A** contains extended related work on Formal XAI.

**Appendix B** contains the formalizations of the models and distributions used in this work.

**Appendix C** contains the proof of Proposition 1.

**Appendix D** contains the proof of Proposition 2.

**Appendix E** contains the proof of Proposition 3.

**Appendix F** contains the proof of Proposition 4.

**Appendix G** contains the proof of Proposition 5.

**Appendix H** contains the proof of Theorem 1.

**Appendix I** contains the proof of Proposition 6.

**Appendix J** contains the proof of Theorem 2.

**Appendix K** contains the proof of Theorem 3.

**Appendix L** contains the proof of Theorem 4.

**Appendix O** contains an LLM usage disclosure.

## A EXTENDED RELATED WORK ON FORMAL XAI

Our work belongs to the line of research on *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanations endowed with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A central theme of this area is the computational complexity of generating such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Since formal explanations are typically intractable to compute for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), much of the literature has concentrated on more tractable model classes (Marques-Silva & Ignatiev, 2023) — decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026).

Beyond complexity-theoretic analysis, considerable effort has been devoted to computing formal explanations in practice via automated reasoning tools — MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) for deriving certified explanations of neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). A complementary body of work seeks to mitigate the underlying computational burden through abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), and training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, a related strand of formal XAI applies combinatorial and submodular optimization techniques to the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), particularly for decision tree classifiers. As that literature observes — and as we confirm here (Appendix F)— the value-function objective underlying these local variants is not, in fact, submodular; greedy algorithms inspired by submodular optimization therefore operate only as heuristics, albeit empirically effective ones. Our contribution here is a precise delineation of where the two settings part company. Whereas that line of work applies submodular machinery to a local objective that lacks submodularity, we prove that the *global* value functions genuinely possess the requisite structure — monotonicity unconditionally, and submodularity or supermodularity under feature independence — and thereby convert the same greedy machinery from a heuristic into an algorithm with formal approximation guarantees. The rigorous coupling of combinatorial-optimization objectives to the *global, probabilistic* variants of sufficient and contrastive reasons is, to our knowledge, new, and it opens the door to a broader family of principled algorithms.

## B MODEL AND DISTRIBUTION FORMALIZATIONS

This section supplies the formal definitions of the models and distributions used throughout the paper: Subsection B.1 defines the model families, and Subsection B.2 the distribution classes.

### B.1 MODEL FORMALIZATIONS

This subsection formalizes the three base model types analyzed throughout the paper: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

**Decision Trees.** A decision tree (DT) is a directed acyclic graph representing a function  $f : \mathbb{F} \rightarrow [c]$ , where  $c \in \mathbb{N}$  is the number of classes, subject to the following structure: (i) each internal node  $v$  is labeled with a distinct input feature from  $\{1, \dots, n\}$ ; (ii) each internal node  $v$  has at most  $k$  outgoing edges, one for each value in  $[k]$  assigned to  $v$ ; (iii) no variable is tested more than once along any root-to-leaf path  $\alpha$ ; (iv) each leaf is labeled with a class in  $[c]$ . An input  $\mathbf{x} \in \mathbb{F}$  thus traces a unique path  $\alpha$  from the root to a leaf, and the output  $f(\mathbf{x})$  is the class label  $i \in [c]$  of that leaf. The size  $|f|$  of the DT is its total number of edges. To accommodate flexible modeling, the variable ordering may differ across distinct paths  $\alpha$  and  $\alpha'$ , subject only to the constraint that no variable recurs within a single path.

**Neural Networks.** We prove our *hardness* results for neural networks with ReLU activations, whereas our *tractability* results (the polynomial-time algorithms) hold for any architecture admitting polynomial-time inference — a standard and mild assumption. Consequently, all of our tractable–intractable separations carry over to *every* neural architecture at least as expressive as a standard feed-forward ReLU network, including such widely used models as ResNets, CNNs, Transformers, and diffusion models. Since any ReLU network can be rendered fully connected by assigning zero weights and biases to the absent connections, we follow standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025) and assume full connectivity, so that our analysis concerns multi-layer perceptrons (MLPs) without loss of generality. Formally, an MLP  $f$  comprises  $t - 1$  *hidden layers*  $(g^{(j)})$  for  $j = 1, \dots, t - 1$  together with an output layer  $g^{(t)}$ , each layer defined recursively by:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where  $W^{(j)}$ ,  $b^{(j)}$ , and  $\sigma^{(j)}$  denote, respectively, the weight matrix, bias vector, and activation function of the  $j$ -th layer. The model is thus specified by  $t$  weight matrices  $(W^{(1)}, \dots, W^{(t)})$ ,  $t$  bias vectors  $(b^{(1)}, \dots, b^{(t)})$ , and  $t$  activation functions  $(\sigma^{(1)}, \dots, \sigma^{(t)})$ .

The input layer is  $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$ , the binary input vector. The layer widths are fixed by a sequence of positive integers  $d_0, \dots, d_t$ , with  $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$  and  $b^{(j)} \in \mathbb{Q}^{d_j}$  the (trained) rational-valued weight matrices and bias vectors, respectively. As the model is a binary classifier over  $n$  features, we set  $d_0 = n$  and  $d_t = 1$ . The hidden layers employ the ReLU activation  $\text{ReLU}(x) = \max(0, x)$ . Although training typically employs a sigmoid output activation, for post-hoc interpretability we take the output layer to apply the threshold step function  $\text{step}(\mathbf{z}) = 1$  if  $\mathbf{z} \geq 0$  and  $\text{step}(\mathbf{z}) = 0$  otherwise.

**Tree Ensembles.** Because our concern is post-hoc explanation, we abstract away the training procedure and specify ensembles purely by their *inference* rule. We treat the two dominant families: ensembles that predict by *majority voting* (e.g., random forests) and those that predict by *weighted voting* (e.g., XGBoost). As with neural networks, our *tractability* results — the polynomial-time algorithms — extend to *any* ensemble configuration with polynomial-time inference, and thus cover a considerably broader range of ensemble designs than these two families alone.

**Majority Voting Inference.** Under *majority voting*, the prediction  $f(\mathbf{x})$  is the class  $j \in [c]$  receiving the most votes among the individual predictions  $f_i(\mathbf{x})$  across all trees  $i \in [k]$  of the ensemble:

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where  $\text{majority}(S)$  returns the most frequent label in the multiset  $S$ , with ties resolved by a fixed rule (e.g., lexicographic order or a predefined priority).

**Weighted Voting Inference.** Under *weighted voting*, each base model  $i$  carries a weight  $\phi_i \in \mathbb{Q}$  encoding its relative importance, and the predicted class is the one accumulating the greatest aggregate weight. Formally, for any  $\mathbf{x} \in \mathbb{F}$ :

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where  $\mathbb{I}$  denotes the identity function.

## B.2 DISTRIBUTION FORMALIZATIONS

This subsection formalizes the distribution classes introduced in the main paper, making precise the distinction — emphasized there — between the empirical and independent classes.

**Empirical Distributions.** An empirical distribution  $\mathcal{D}$  over the input features is induced by a dataset  $\mathbf{D}$  of inputs  $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$ , assigning to each  $\mathbf{x} \in \mathbb{F}$  a probability equal to its frequency of occurrence in  $\mathbf{D}$ :

$$\mathbf{Pr}(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

**Independent Distributions.** Given a marginal probability  $p(\mathbf{x}_i) \in [0, 1]$  for each individual feature, the distribution  $\mathcal{D}$  is *independent* if the joint probability factorizes as a product of marginals,  $\mathbf{Pr}(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$ . This factorization is precisely the property our supermodularity and submodularity proofs invoke; it is a structural assumption on  $\mathcal{D}$ , not a consequence of any dataset. In particular, an empirical distribution places its mass on the observed points of  $\mathbf{D}$  and is product-distributed only in the degenerate case where those points themselves factorize — so the two classes coincide only exceptionally, and a result stated for one does not transfer to the other by default.

When  $p(\mathbf{x}_i) = p(\mathbf{x}_j)$  for all  $i, j \in [n]$ , this reduces to the *uniform distribution*, itself a special case of the independent distributions.

**General Distributions.** Many of our proofs — in particular those establishing the intrinsic monotonicity of the value functions — hold under fully general distributional assumptions, invoking neither independence nor an empirical form. While empirical distributions (the training dataset used as a proxy) are the most common choice in XAI, a range of alternative frameworks exists for approximating the input distribution:  $k$ -NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for modeling tabular dependencies (Größer & Okhrin, 2022), and more expressive conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022). These choices are standard in the counterfactual-explanation and algorithmic-recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which is itself closely tied to contrastive explanations. Moreover, when the value function admits neither a direct structural computation (e.g., leaf enumeration in decision trees) nor an exact empirical evaluation, it can be approximated by Monte Carlo sampling (Hastings, 1970), along the lines of (Subercaseaux et al., 2025; Lopardo et al., 2023); characterizing when such approximation preserves the minimality guarantees is one of the open directions noted in the main text.

## B.3 SUBSET VS. CARDINAL MINIMALITY

This subsection develops the distinction between cardinal and subset minimality in greater detail. Cardinal minimality is the substantially stronger of the two guarantees: it corresponds to globally minimal explanation size, whereas subset minimality guarantees only a local form of minimality — irreducibility under single-feature removal.

To see why cardinal minimality is stronger, suppose  $S \subseteq [n]$  is a cardinally minimal explanation. Then no set  $S'$  with  $|S'| < |S|$  is an explanation; in particular no strict subset  $S' \subsetneq S$  is an explanation, so  $S$  is subset minimal as well.

The converse, however, fails. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

together with the assignment  $\mathbf{x} := (1, 1, 1)$ , for which  $f(1, 1, 1) = 1$ . Fixing  $\mathbf{x}_1 = 1$  yields a cardinally minimal (and hence subset-minimal) sufficient reason, since the prediction remains 1 under every assignment of  $\mathbf{x}_2, \mathbf{x}_3$ . Fixing both  $\mathbf{x}_2 = 1$  and  $\mathbf{x}_3 = 1$  likewise yields a subset-minimal explanation — yet not a cardinally minimal one, as it is strictly larger than  $\{1\}$ . Thus every cardinally minimal explanation is subset-minimal, but not conversely, which is exactly why the two notions call for separate complexity treatments.

## C PROOF OF PROPOSITION 1

**Proposition 1.** *While the non-probabilistic case ( $\delta = 1$ ) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary  $\delta$ ), there exist functions  $f$  that have  $\Theta(\frac{2^n}{\sqrt{n}})$  subset-minimal global sufficient reasons.*

*Proof.* It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input  $\mathbf{x}$  Bassan et al. (2024), a phenomenon realized by functions of the form below. We establish that the very same function admits an exponential number of *global* and *probabilistic* sufficient reasons. This function is a special case of the broader class of *threshold Boolean functions* of Wegener (2005): for  $n = 2k + 1$  with  $k \in \mathbb{N}$ , it is defined by:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where  $\mathbf{x}$  is drawn from a uniform distribution  $\mathcal{D}$ . The function  $f$  is *symmetric* in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends only on the number of 1's (equivalently, 0's) in the input and is invariant under every permutation of the input bits. Moreover, the two extreme inputs — the all-zeros vector  $(0, \dots, 0)$  and the all-ones vector  $(1, \dots, 1)$  — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons, and it is this combinatorial abundance that the argument below transfers to the global probabilistic setting.

Since  $f$  is symmetric, the identity  $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$  holds for any two subsets  $S_1, S_2 \subseteq [n]$ . Furthermore, one verifies that for every subset  $S \subseteq [n]$  of size  $k + 1$ :

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set  $\delta$  to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and set  $k := \lfloor \frac{n}{2} \rfloor$ . Each such subset is *subset-minimal*, since any subset of size at most  $\lfloor \frac{n}{2} \rfloor - 1$  fails to be a sufficient reason for  $\langle f, \mathbf{x} \rangle$  at the threshold  $\delta$ . The analysis of Bassan et al. (2024) therefore applies verbatim, yielding exactly  $\binom{n}{\lfloor \frac{n}{2} \rfloor}$  subset-minimal local sufficient reasons for  $\langle f, \mathbf{x} \rangle$ . By Stirling's approximation:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This establishes the claimed  $\Theta(\frac{2^n}{\sqrt{n}})$  bound on the number of subset-minimal global  $\delta$ -sufficient reasons, confirming that uniqueness fails dramatically once  $\delta < 1$ .

□

## D PROOF OF PROPOSITION 2

**Proposition 2.** *While the local probabilistic setting (for any  $\delta$ ) lacks monotonicity — i.e., the value functions  $v_{con}^l$  and  $v_{suff}^l$  are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any  $\delta$ ), both value functions  $v_{con}^g$  and  $v_{suff}^g$  are monotonic non-decreasing.*

*Proof.* We prove monotonicity for both global explanation types, sufficient and contrastive. We treat global sufficiency first, building intuition through the Boolean case and then generalizing successively to functions with discrete multi-valued input domains and multiple output classes, and thereafter to continuous input domains — thereby covering every well-defined classification function. We then derive the corresponding guarantee for global contrastive explanations by duality. Crucially, none of these arguments invokes feature independence: monotonicity holds in the strong sense of being valid for *any* underlying distribution, which is what makes it, rather than the modularity properties, the workhorse of our subset-minimal tractability results.

Finally, we exhibit a simple example demonstrating that monotonicity fails for the local probabilistic sufficient and contrastive value functions, confirming that the property is genuinely a global phenomenon rather than an artifact of the probabilistic relaxation.

**Lemma 1.** *The global sufficient value function  $v_{suff}^g$  is monotonic non-decreasing for Boolean functions, under any data distribution.*

*Proof.* Given an arbitrary set  $S \in [n]$ , and a feature  $i \notin S$ , we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points  $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$  such that  $\mathbf{x}_S = \mathbf{x}'_S$  and  $f(\mathbf{x}) = 1 - f(\mathbf{x}')$ , it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

For notational economy, we occasionally suppress the distribution subscript  $\sim \mathcal{D}$  in the local probability  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$ , and we write  $f^+$  and  $f^-$  for the events  $f(\mathbf{z}) = 1$  and  $f(\mathbf{z}) = 0$ , respectively.

The argument proceeds in two stages: we first carry out a technical simplification of the probability term, and then establish monotonicity from the simplified expression.

**Simplifying the probability term.** Let  $\mathcal{D}_S$  denote the distribution restricted to  $S$ . Fix a point  $\mathbf{x} \sim \mathcal{D}$  and a set  $S$ , and consider all points  $\mathbf{z} \sim \mathcal{D}$  with  $\mathbf{z}_S = \mathbf{x}_S$  and  $f(\mathbf{z}) = b$  for a fixed  $b \in \{0, 1\}$ ; all of these share the same local probability  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$ . Marginalizing over  $\mathbf{x}_{\bar{S}}$  across every  $\mathbf{x} \sim \mathcal{D}$  with the same  $\mathbf{x}_S$  and the same output  $b$  yields  $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$ , whence:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Consequently, for the expectation  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$  we obtain:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2],
\end{aligned} \tag{22}$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

**Proving monotonicity.** Building on the preceding simplification, we now establish monotonicity. We introduce additional notation: let  $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$  and  $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$ , and let  $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$  and  $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$ . The two expectations  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$  and  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$  then simplify accordingly<sup>1</sup>:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = 0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

<sup>1</sup>The notations  $P_1$ ,  $P_0$ ,  $g_1^+$ , and  $g_0^+$  — like all others used throughout the paper — are not fixed constants but depend on the specific input  $\mathbf{x}_S$ ; formally they should be written  $P_1(\mathbf{x}_S)$ ,  $P_0(\mathbf{x}_S)$ , and so on, but we omit this explicit dependence for readability.

Let  $\Delta_t(S, f)$  denote  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ . Without loss of generality, we assume that both  $P_1$  and  $P_0$  are positive. Otherwise, it can be verified that  $\Delta_t(S, f) = 0$ . Then, we get:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \mathbf{Pr}(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\ &\geq 0. \end{aligned} \tag{26}$$

Note that  $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$  and, symmetrically,  $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$ . Combining these identities with the preceding computation yields  $\Delta_t(S, f) \geq 0$ , so  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$  is monotone non-decreasing, which completes the proof in the Boolean case.  $\square$

**Lemma 2.** *The global sufficient value function  $v_{\text{diff}}^g$  is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

*Proof.* Extending the Boolean argument, we now generalize to functions whose input domains and output ranges are each finite sets of values. Let  $\mathcal{K}$  denote the set of output classes. The structure of the derivation is unchanged; only the bookkeeping over multiple values and classes grows.

Equation 22 generalizes to:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] \tag{27}$$

Conditioning on  $\mathbf{z}_S = \mathbf{x}_S$ , and letting feature  $t$  range over its  $r$  possible values  $\{v_1, \dots, v_r\}$ , equation 23 generalizes to:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \sum_{k=1}^r \left[ \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right]. \end{aligned} \tag{28}$$

Let  $P_k$  denote  $\mathbf{Pr}(z_t = v_k \mid \mathbf{x}_S)$ , Let  $g_k^j$  denote  $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$ , that is,  $z_t$  takes the value  $v_k$  and the output class is  $j$ . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[ \sum_{k=1}^r P_k \cdot \left( \sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{k=1}^r P_k \cdot \left( \sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left( \sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{30}$$

Since  $\Delta_t(S, f) \geq 0$ , the value function  $\mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$  is monotone non-decreasing, completing the proof in the discrete multi-valued setting.  $\square$

**Lemma 3.** *The global sufficient value function  $v_{\text{diff}}^g$  is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

*Proof.* We now extend monotonicity to features defined over continuous domains. One route partitions the continuous domain into two disjoint subdomains and adapts the proof of lemma 2 by relabeling notation accordingly; because both summation and integration are linear operators, the core reasoning transfers unchanged from the discrete to the continuous case, and only the notation differs.

For completeness, we instead give a direct proof. Assume every feature lies in a continuous domain, and let  $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$  denote the joint probability density,  $p(x_i)$  the marginal density of  $x_i$ , and  $\mathcal{X}_i$  the domain of feature  $i$ .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \tag{31}$$

where  $\mathcal{X}_{\bar{S}}$  is the domain of the features in  $\bar{S}$ , and the conditional density as:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \tag{32}$$

The local probability of  $S$  given  $\mathbf{x}$  is

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \tag{33}$$

which, via an additional feature  $t \notin S$ , decomposes as:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \tag{34}$$

We then modify equation 27 to

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S,
\end{aligned} \tag{35}$$

while equation 28 becomes

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left( \sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For brevity, write  $Y_j(t) := \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$  and  $p_{t|S} := p(z_t \mid \mathbf{x}_S)$ , both of which are associated with a fixed  $\mathbf{x}_S$ <sup>2</sup>. Monotonicity then follows from:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left( \sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \left( \int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left( \int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

**Proposition 2.** While the local probabilistic setting (for any  $\delta$ ) lacks monotonicity — i.e., the value functions  $v_{con}^\ell$  and  $v_{suff}^\ell$  are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any  $\delta$ ), both value functions  $v_{con}^g$  and  $v_{suff}^g$  are monotonic non-decreasing.

*Proof.* Combining lemmas 1 to 3, the global sufficient value function  $v_{suff}^g$  is monotonic non-decreasing for every well-defined classification function — whether its features are Boolean, discrete, continuous, or a mixture thereof — under any data distribution. It remains only to transfer this to the contrastive setting.

$$v_{con}^g(S) = 1 - v_{suff}^g(\bar{S}) \quad (38)$$

Hence  $v_{con}^g$  is monotonic non-increasing as features are added to  $\bar{S}$  or, equivalently, monotonic non-decreasing as features are removed from  $\bar{S}$  and added to  $S$ , which completes the proof.

□

## E PROOF OF PROPOSITION 3

**Proposition 3.** While the local probabilistic sufficient setting (for any  $\delta$ ) lacks supermodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{suff}^\ell$  is not supermodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{suff}^g$  is supermodular.

*Proof.* This section establishes supermodularity for the global sufficient (and, by duality, contrastive) setting. As with monotonicity, we proceed from Boolean functions through multi-valued discrete

<sup>2</sup>Changing  $\mathbf{x}_S$  alters their values; we suppress  $\mathbf{x}_S$  in the notation for brevity.

and continuous input domains to the general well-defined classification function. In contrast to monotonicity, however, these proofs *require* the assumption of feature independence, and we exhibit a counterexample confirming that supermodularity can fail once this assumption is relaxed — so the independence hypothesis is not merely convenient but necessary.

**Lemma 4.** *The global sufficient value function  $v_{\text{diff}}^g$  is supermodular for Boolean functions, under the assumption of feature independence.*

*Proof.* Concretely, we prove that

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

for all  $S \subseteq S'$  with  $t \notin S'$  — that is, that the marginal gain  $\Delta_t(S, f)$  is itself monotone in  $S$ , which is exactly supermodularity.

The case  $S' = S$  is immediate. It therefore suffices to treat  $S' = S \cup \{i\}$  with  $i \notin S$  and  $i \neq t$ . Under feature independence, the local probability  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$  satisfies:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \\ = \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Write  $P_1, P_0, Q_1$ , and  $Q_0$  for  $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$ , and  $\Pr(z_i = 0)$ , respectively, and let  $g_{11}^+, g_{10}^+, g_{01}^+$ , and  $g_{00}^+$  denote  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$ , and  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$ , respectively. We then decompose equation 26 as:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

Since this difference is non-negative,  $\Delta_t(S, f)$  is monotone — equivalently,  $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$  — establishing supermodularity.  $\square$

**Lemma 5.** *The global sufficient value function  $v_{\text{diff}}^g$  is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

*Proof.* We extend the preceding argument to multi-valued discrete functions, again under feature independence. Write  $P_k := \Pr(z_t = v_k), Q_{k'} := \Pr(z_i = v_{k'}')$ , and  $g_{k'k}^j := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}', z_t = v_k)$ . Decomposing equation 30 then gives

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k P_l \cdot \left( \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k P_l \cdot \left( \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right)^2 \right]. \end{aligned} \quad (44)$$

Likewise, we have

$$\begin{aligned}
\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[ \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v_k^i}} \Pr(\mathbf{x}_S) \cdot \Pr(z_i = v_k^i) \cdot \sum_{j \in \mathcal{K}} \left( \sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[ \sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left( \sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].
\end{aligned} \tag{45}$$

Therefore, we get

$$\begin{aligned}
&\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[ \sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{46}$$

□

**Proposition 3.** *While the local probabilistic sufficient setting (for any  $\delta$ ) lacks supermodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{diff}}^\ell$  is not supermodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{diff}}^g$  is supermodular.*

*Proof.* We extend the supermodularity result of Lemma 5 to the continuous case, assuming that  $\mathcal{D}_p$  satisfies feature independence. One route partitions the continuous domain into two disjoint subdomains and adapts the proof of Lemma 5 by relabeling notation.

For completeness, we give a direct proof. Under feature independence, the marginal and conditional densities simplify to:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \tag{47}$$

Consequently, the local probability of  $S$  conditioned on  $\mathbf{x}$  reduces to:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \tag{48}$$

Write  $Y_j(i, t) := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$ , where  $i \notin S, t \notin S$  and  $i \neq t$ . Then clearly

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
&\Delta_t(S, f) \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_i} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left( \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left( \int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left( \int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\ &= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}. \end{aligned}$$

Let  $S' = S \cup \{i\}$ , then we have:

$$\begin{aligned} & \Delta_t(S', f) \\ &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \quad (50) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[ \sum_{j \in \mathcal{K}} \frac{1}{2} \left( \int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S, \end{aligned}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\ &= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}. \end{aligned}$$

Let  $\alpha(i, t, t')$  denote  $Y_j(i, t) - Y_j(i, t')$  and  $\beta(i', t, t')$  denote  $Y_j(i', t) - Y_j(i', t')$ , we can infer that

$$\begin{aligned} & \Delta_t(S', f) - \Delta_t(S, f) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[ \sum_{j \in \mathcal{K}} \frac{1}{4} \left( \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (51)$$

Since this difference is non-negative, the global sufficient value function  $v_{\text{suff}}^g$  retains supermodularity for every valid classification function, whether its features are Boolean, discrete, continuous, or a mixture thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables  $\{x_1, x_2, x_3\}$  and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \quad (52)$$

Consider the input  $\mathbf{x} = (1, 1, 1)$  which is classified as 1. For the local probabilistic sufficient value function  $v_{\text{suff}}^l$ , we have that the following holds:

$$\begin{aligned} v_{\text{suff}}^l(\emptyset) &= \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4}, \\ v_{\text{suff}}^l(\{1, 2\}) &= \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1. \end{aligned}$$

Since  $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$ , the function  $v_{\text{suff}}^l$  is neither supermodular nor submodular. □

## F PROOF OF PROPOSITION 4

**Proposition 4.** *While the local probabilistic contrastive setting (for any  $\delta$ ) lacks submodularity — even when  $\mathcal{D}$  is uniform, i.e., the value function  $v_{\text{con}}^l$  is not submodular — in the global probabilistic setting (also for any  $\delta$ ), when  $\mathcal{D}$  exhibits feature independence, the value function  $v_{\text{con}}^g$  is submodular.*

*Proof.* Once we establish that the global sufficient setting is supermodular, it directly follows that the global contrastive setting is submodular. Let  $S \subseteq [n]$  be the set of features allowed to change, and suppose  $S \subseteq S'$  with  $i \notin S'$ . To compare the marginal contributions  $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$  and  $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$ , we equivalently compare  $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$  and  $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$ . Since  $S \subseteq S'$ , it follows that  $\bar{S}' \subseteq \bar{S}$ , and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function  $v_{\text{con}}^g$  is submodular, hence concluding this proof.

We will now demonstrate that the local probabilistic contrastive value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables  $\{x_1, x_2, x_3\}$  and define a Boolean function as follows:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input  $\mathbf{x} = (1, 1, 1)$  which is classified as 1. For the local probabilistic contrastive value function  $v_{\text{con}}^l$ , we have:

$$\begin{aligned} v_{\text{con}}^l(\emptyset) &= 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2}, \\ v_{\text{con}}^l(\{1, 2\}) &= \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}. \end{aligned}$$

Since  $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$ , the function  $v_{\text{con}}^l$  is neither submodular nor supermodular. □

In the rest of this section, we show that the feature independence assumption is not just sufficient but also necessary for supermodularity. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this boolean function, it holds that  $f(00) = 0$ ,  $f(01) = 0$ ,  $f(10) = 1$ , and  $f(11) = 1$ . Moreover, define a distribution  $\mathcal{D}$  on  $x_1$  and  $x_2$ . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where  $S \subseteq S'$  and  $t \notin S'$ , then the global sufficient value function is supermodular. When  $S = \emptyset$ , we have that the following holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) \\
& = (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2
\end{aligned}$$

When  $S = \{2\}$ , we have that the following condition holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1)
\end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)}
\end{aligned}$$

In contrast, when  $S = \{1\}$ , we have the the following holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1)
\end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when  $S = \{1, 2\}$ , we have the following condition:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0)} + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1)} + \\
& \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 1, x_2 = 0)} + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= 1
\end{aligned}$$

Let  $t = 2$ . Evidently, we have  $\Delta_t(\{1\}, f) = 0$ . Next, let us analyse  $\Delta_t(\emptyset, f)$ . We note that if we assume feature independence, we observe that:

$$\begin{aligned}
& (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned}
& \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2.
\end{aligned}$$

This implies  $\Delta_t(\emptyset, f) = 0$ , that is,  $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$ . However, without the assumption of feature independence, we cannot guarantee  $\Delta_t(\emptyset, f) = 0$ . For example, let:

$$\mathbf{Pr}(x_1 = 0, x_2 = 0) = P_{00}, \mathbf{Pr}(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\mathbf{Pr}(x_1 = 1, x_2 = 0) = P_{10}, \mathbf{Pr}(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying  $\Delta_t(\emptyset, f)$  we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[ \frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[ (P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let  $P_{00} = 0.4$ ,  $P_{01} = 0.2$ ,  $P_{10} = 0.1$ , and  $P_{11} = 0.3$ , and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[ \frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[ (0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that  $\Delta_t(\emptyset, f) > \Delta_t(1, f)$ , indicating that — unlike in the monotonicity setting where it was not required — feature independence is a necessary condition for supermodularity.

## G PROOF OF PROPOSITION 5

**Proposition 5.** *Computing Algorithm 1 with the local value functions  $v_{con}^\ell$  and  $v_{suff}^\ell$  does not always converge to a subset minimal  $\delta$ -sufficient/contrastive reason. However, computing it with the global value functions  $v_{con}^g$  or  $v_{suff}^g$  necessarily produces subset minimal  $\delta$ -sufficient/contrastive reasons.*

*Proof.* We note that this proposition follows directly from our proof that both  $v_{\text{suff}}^g$  and  $v_{\text{con}}^g$  are monotone non-decreasing (as established in Proposition 2), along with other relevant conclusions drawn in prior work. In the remainder of this section, we elaborate on how this monotonicity property leads to the stated corollary.

The convergence of Algorithm 1 to a subset-minimal  $\delta$  global sufficient or contrastive reason with respect to the value functions  $v_{\text{con}}^g$  and  $v_{\text{suff}}^g$  follows directly from the monotonicity property, for which prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) showed that this type of greedy algorithm yields a subset-minimal explanation when the underlying value function is monotone non-decreasing. The algorithm halts when, for all  $i \in S$ , the value  $v(S \setminus \{i\})$  drops below  $\delta$ , ensuring that although  $v(S) \geq \delta$  (a maintained invariant), removing any element causes the condition to fail. Due to monotonicity, this implies that no proper subset of  $S$  satisfies the condition, guaranteeing that  $S$  is subset-minimal.

We now turn to explain why Algorithm 1 does not converge to a subset-minimal solution in the local setting. We note that it is well known that the local probabilistic sufficient value function (and likewise the local probabilistic contrastive value function) is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution on the two binary variables  $\{x_1, x_2\}$  and define a Boolean function as follows:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input  $\mathbf{x} = (0, 1)$  which is classified as 1, then:

$$v_{\text{suff}}^\ell(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^\ell(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^\ell(\{2\}) = 1, \quad v_{\text{suff}}^\ell(\{1, 2\}) = 1. \quad (60)$$

Because  $v_{\text{suff}}^\ell(\emptyset) > v_{\text{suff}}^\ell(\{1\}) < v_{\text{suff}}^\ell(\{1, 2\})$ , the function  $v_{\text{suff}}^\ell$  is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^\ell(\emptyset) = 1, \quad v_{\text{con}}^\ell(\{1\}) = 1, \quad v_{\text{con}}^\ell(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^\ell(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For  $v_{\text{con}}^\ell$ , the  $S$  denotes the set of features that are allowed to vary.) Because  $v_{\text{con}}^\ell(\emptyset) > v_{\text{con}}^\ell(\{2\}) < v_{\text{con}}^\ell(\{1, 2\})$ , the function  $v_{\text{con}}^\ell$  is also non-monotone.

□

## H PROOF OF THEOREM 1

**Theorem 1.** *If  $f$  is a decision tree and the probability term  $v_{\text{suff}}^g$  can be computed in polynomial time given the distribution  $\mathcal{D}$  (which holds for independent distributions, among others), then obtaining a subset-minimal global  $\delta$ -sufficient reason can be obtained in polynomial time. However, unless  $\text{PTIME}=\text{NP}$ , no polynomial-time algorithm exists for computing a local  $\delta$ -sufficient reason for decision trees even under independent distributions.*

*Proof.* We begin by noting that for decision trees, assuming that  $\mathcal{D}$  represents independent distributions, the computation of the value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

Can be carried out in polynomial time using the following procedure. Thanks to the tractability of decision trees, we can iterate over all pairs of leaf nodes, each corresponding to partial assignments  $\mathbf{x}_S$  and  $\mathbf{z}_S$ . For each such pair, if both leaf nodes yield the same prediction under  $f$ , we compute the corresponding term in the expected value  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$  by multiplying the respective feature-wise probabilities over the shared features of the two vectors. Under the feature independence assumption, these probabilities decompose, allowing the full expectation to be computed by summing the contributions of all such matching leaf pairs.

Since all probabilities involved are provided as part of the input and each step involves only polynomial-time operations, the entire procedure runs in polynomial time. This establishes that every step in Algorithm 1 is efficient, and thus the algorithm as a whole runs in polynomial time. Finally, combining this with Lemma 2, which proves that  $v_{\text{suff}}^g$  is monotone non-decreasing, we conclude that Algorithm 1 converges to a subset-minimal explanation. This is because the algorithm halts when, for all  $i \in S$ , the value  $v(S \setminus \{i\})$  falls below  $\delta$ , while  $v(S) \geq \delta$  is preserved throughout. By monotonicity, this guarantees that no strict subset of  $S$  satisfies the condition, ensuring  $S$  is indeed subset-minimal.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which showed that, assuming  $P \neq NP$ , there is no polynomial-time algorithm for computing a subset-minimal  $\delta$ -sufficient reason for decision trees. While this was proven specifically under the uniform distribution, the hardness clearly extends to independent distributions, which include this case. Combined, these results establish the complexity separation between the local and global variants, thus concluding our proof.

□

## I PROOF OF PROPOSITION 6

**Proposition 6.** *For any model  $f$ , and empirical distribution  $\mathcal{D}$  — computing a subset-minimal global  $\delta$ -sufficient or  $\delta$ -contrastive reason for  $f$  can be done in polynomial time.*

*Proof.* The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

Can be performed in polynomial time when  $\mathcal{D}$  is selected from the class of empirical distributions. This is achieved by iterating over pairs of instances  $\mathbf{x}, \mathbf{z}$  within the dataset and running an inference through  $f(\mathbf{x})$  and  $f(\mathbf{z})$  to compute the expected values at each step. Consequently, determining whether the probability function exceeds or falls below a given threshold  $\delta$  can also be accomplished in polynomial time. Furthermore, leveraging the proof of non-decreasing monotonicity (Proposition D) of the global probability function, we can utilize algorithm 1 for contrastive reason search or for sufficient reason search. By executing a linear number of polynomial queries, a subset-minimal explanation (whether a subset minimal contrastive reason or a subset minimal sufficient reason) can thus be obtained in polynomial time.

□

## J PROOF OF THEOREM 2

**Theorem 2.** *Assuming  $f$  is a neural network or a tree ensemble, and  $\mathcal{D}$  is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global  $\delta$ -sufficient and contrastive reasons. However, unless  $PTIME=NP$ , there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -sufficient reason or a subset minimal local  $\delta$ -contrastive reason.*

*Proof.* The first part of the proof follows directly from Proposition 6, which established that this holds for any model, since both  $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$  and  $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$  can be computed in polynomial time from the empirical distribution. For the second part, we proceed by proving two lemmas: one establishing the intractability of computing the sufficient case, and the other addressing the contrastive case.

**Lemma 6.** *Unless  $PTIME=NP$ , then there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

*Proof.* We will first prove this claim for neural networks and then extend the result to tree ensembles. We will establish this claim by demonstrating that if a polynomial-time algorithm exists for computing a subset-minimal local  $\delta$  sufficient reason for neural networks, then it would enable us to solve the classic NP-complete *CNF-SAT* problem in polynomial time. The *CNF-SAT* problem is defined as follows:

**CNF-SAT:**

**Input:** A formula in conjunctive normal form (CNF):  $\phi$ .

**Output:** Yes, if there exists an assignment to the  $n$  literals of  $\phi$  such that  $\phi$  is evaluated to True, and No otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

**Lemma 7.** Any boolean circuit  $\phi$  can be encoded into an equivalent MLP over the binary domain  $\{0, 1\}^n \rightarrow \{0, 1\}$  in polynomial time.

*Proof.* We will actually establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution  $\mathcal{D}$  consists of a single element, which, for simplicity, we take to be the zero vector  $\mathbf{0}_n$ . Since the sufficiency of  $S$  in this scenario depends only on the local instance  $\mathbf{x}$  and the single instance  $\mathbf{0}_n$  in our empirical distribution, the nature of the input format (whether discrete, continuous, etc.) does not affect the result. Therefore, the hardness results hold universally across all these settings.

Similarly to  $\mathbf{0}_n$ , we define  $\mathbf{1}_n$  as an  $n$ -dimensional vector consisting entirely of ones. Given an input  $\phi$ , we initially assign  $\mathbf{1}_n$  to  $\phi$ , effectively setting all variables to True. If  $\phi$  evaluates to True under this assignment, then a satisfying truth assignment exists. Therefore, we assume that  $\phi(\mathbf{1}_n) = 0$ . We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

$\phi'$ , while no longer a CNF, can still be transformed into an MLP  $f$  using Lemma 8, ensuring that  $f$  behaves equivalently to  $\phi'$  over the domain  $\{0, 1\}^n$ . Given our assumption that computing a subset-minimal local  $\delta$ -sufficient reason is feasible in polynomial time, we can determine one for the instance  $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$ ,  $\delta := 1$ , noting that the empirical distribution  $\mathcal{D}$  is simply defined over the single data point  $\mathbf{0}_n$ .

We now assert that the subset-minimal  $\delta$ -sufficient reason generated for  $\langle f, \mathbf{x} \rangle$  encompasses the entire input space, i.e.,  $S = \{1, \dots, n\}$ , if and only if  $\langle \phi \rangle \notin \text{CNF-SAT}$ .

Let us assume that  $S = \{1, \dots, n\}$ . Since  $S$  is a  $\delta$ -sufficient reason for  $\langle f, \mathbf{x} \rangle$ , this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is  $\emptyset$  in this case, this trivially holds. The fact that  $S$  is subset-minimal means that any other subset  $S' \subseteq S$  satisfies  $v(S') < \delta = 1$ . Since the probability function  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$  is determined by a single point (the distribution contains only the point  $\mathbf{0}_n$ ), the probability function can only take values of 1 or 0. Hence, we also know that  $v(S') = 0$ . This tells us that, aside from the subset  $S = \{1, \dots, n\}$ , for any subset  $S' \subseteq S$ , fixing the features in  $S'$  to 1 and the rest to 0 does not result in a classification outcome of 1. Since the  $\phi_2$  component within  $\phi'$  is True only if all features are assigned 1, this directly implies that  $\phi$  is assigned False for any of these inputs. Since we already know that  $\phi$  does not return a True answer for the vector assignment  $\mathbf{1}_n$  (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that  $\phi \notin \text{CNF-SAT}$ .

Now, suppose that  $S$  is a subset that is strictly contained within  $\{1, \dots, n\}$ . Given that  $S$  is sufficient under our definitions of the distribution  $\mathcal{D}$  with  $\delta = 1$ , we can apply the same reasoning as before to conclude that  $v(S) = 1$ . This implies that setting the features in  $S$  to 1 while setting the remaining features to 0 ensures that the function  $f$  evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the  $\phi_2$  component within  $\phi'$  must be False. Consequently, the  $\phi$  component must be True, which implies that  $\langle \phi \rangle \in \text{CNF-SAT}$ . This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

**Lemma 8.** *Any CNF or DNF  $\phi$  can be encoded into an equivalent random forest classifier over the binary domain  $\{0, 1\}^n \rightarrow \{0, 1\}$  in polynomial time.*

*Proof.* We observe that we can apply the same process used in our proof for neural networks, where we encoded  $\phi'$  into an equivalent neural network. However,  $\phi'$  is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since  $\phi'$  consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

Where each  $c_i$  is a disjunction of a few terms. Consequently,  $\phi'$  is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

**Lemma 9.** *Unless  $PTIME = NP$ , then there is no polynomial time algorithm for computing a subset minimal local  $\delta$ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

*Proof.* We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula  $\phi$ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of  $\phi$ .

First, we check whether assigning all variables in  $\phi$  to 1 evaluates  $\phi$  to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network  $f$ . Next, we compute a subset-minimal  $\delta$ -sufficient reason by setting  $\mathcal{D}$  as the empirical distribution containing only a single data point  $\mathbf{0}_n$ , following a similar procedure to Lemma 6. Additionally, we set  $\delta = 1$  and compute a subset-minimal  $\delta$ -contrastive reason concerning  $\langle f, \mathbf{x} \rangle$ .

We will now demonstrate that if any subset-minimal  $\delta$  contrastive reason obtained for  $\phi$  is satisfiable, then  $\phi$  itself is satisfiable. Conversely, if no subset-minimal  $\delta$  contrastive reason is obtained, then  $\phi$  is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ , where the distribution  $\mathcal{D}$  considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a  $\delta = 1$  contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of  $f$ , which corresponds to modifying the assignment of  $\phi$  from False to True. If such an assignment exists, then  $\phi$  is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates  $f$  (and consequently  $\phi$ ) to True, implying that  $\phi$  is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

## K PROOF OF THEOREM 3

**Theorem 3.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by  $\mathcal{O}(\ln(|D|))$ , for computing a global cardinally minimal  $\delta$ -contrastive reason for  $f$ , assuming feature*

independence. In contrast, unless  $PTIME=NP$ , no bounded approximation exists for computing a local cardinally minimal  $\delta$ -contrastive reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .

*Proof.* We divide the proof of the theorem into two lemmas, covering both the approximation guarantee and the result on the absence of a bounded approximation.

**Lemma 10.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by  $\mathcal{O}(\ln(|D|))$ , for computing a global cardinally minimal  $\delta$ -contrastive reason for  $f$ , assuming feature independence.*

*Proof.* We will in fact prove a stronger claim, showing that this holds for *any* model, provided the trivial condition that its inference time is computable in polynomial time, along with one additional mild condition that we will detail later — both of which apply to both our neural network and tree ensemble formalizations.

We begin by noting that, since we are working with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs  $\mathbf{x}, \mathbf{z}$  in the dataset  $D$ , as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

Regarding the approximation, the classical work by Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions  $v$  with integer values. More generally, their result yields an approximation factor of  $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$ . We showed in Proposition 2 that the value function  $v_{\text{con}}^g$  is monotone non-decreasing, and under feature independence, Proposition 4 establishes its submodularity. Combined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ .

However, we must also show that the expression  $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$  is both finite and bounded by  $\mathcal{O}(\ln(|D|))$ , implying that it is effectively constant, due to the assumption of a fixed dataset  $|D|$ . To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator  $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$  to be zero. We begin by formally defining what we mean by redundancy:

**Definition 1.** *Let  $\mathcal{D}$  denote some empirical distribution over a dataset  $D$ . Then we say that some feature  $i \in [n]$  is redundant with respect to  $\mathcal{D}$  if for any pair  $\mathbf{x}, \mathbf{z} \in D$  it holds that  $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$ .*

Here, the notation  $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$  indicates that all features in  $[n] \setminus \{i\}$  are fixed to their values in  $\mathbf{x}$ , while feature  $i$  is set to its value in  $\mathbf{z}$ . Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation  $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$  indicates that the features in  $S$  are fixed to their values in  $\mathbf{x}$ , while the features in  $\bar{S}$  are fixed to their values in  $\mathbf{z}$ . Thus, in this sense, if we “remove” feature  $i$  from the input space and define a new function  $f'$  over the reduced space  $[n'] := [n] \setminus \{i\}$ , then for any input of size  $n'$ , the output of  $f'$  will exactly match the output of  $f$  when applied to the same features  $[n] \setminus \{i\}$ . This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it involves detaching the corresponding input neuron from the network; for tree ensembles, it involves removing any splits on that feature from all decision trees. Given the empirical nature of the contrastive value function  $v_{\text{con}}^g(S)$  — as previously discussed in this proof and in Proposition 2 — we can compute each  $v_{\text{con}}^g(\{i\})$  for all  $i \in [n]$  by iterating over all pairs  $\mathbf{x}, \mathbf{z} \in D$  and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature  $i$  is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then  $i$  is deemed redundant. Once all redundant features have been identified with respect to the empirical distribution  $\mathcal{D}$ , we can remove them and construct an equivalent model  $f'$ . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that  $v_{\text{con}}^g(\{i\})$  equals 0 for some empirical distribution  $\mathcal{D}$  if and only if feature  $i$  is redundant with respect to  $\mathcal{D}$ . Therefore, once the preprocessing step removes all redundant features from  $f$  and we construct the resulting function  $f'$ , we ensure that  $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$ .

We now present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is  $\mathcal{O}(\ln(|\mathcal{D}|))$ . Given that the empirical distribution  $\mathcal{D}$  is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish that the bound is *constant* — a property that will later sharply contrast with the local explainability setting, where no bounded approximation exists.

More specifically, we know that the probability function  $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$  is, by the definition of empirical distributions, at least  $\frac{1}{|\mathcal{D}|}$ . Therefore, the definition of  $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$  (i.e., the value function  $v_{\text{con}}^g$ ) is at least  $\frac{1}{|\mathcal{D}|^2}$ . In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|\mathcal{D}|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left( \frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|\mathcal{D}|^2) = \mathcal{O}(\ln(|\mathcal{D}|)) \quad (71)$$

Which concludes our proof.

□

**Lemma 11.** *Unless  $\text{PTIME}=\text{NP}$ , there is no bounded approximation for computing a cardinally minimal local  $\delta$ -contrastive reason for any  $\langle f, \mathbf{x} \rangle$  where  $f$  is either a neural network or a tree ensemble, even when  $|D| = 1$ .*

*Proof.* Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local  $\delta$ -contrastive reason for some  $\langle f, \mathbf{x} \rangle$ , where  $f$  is a neural network or a tree ensemble—even when  $|D| = 1$ . However, Lemma 9 establishes that even with a single baseline  $\mathbf{z} = 0_n$  (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless  $\text{PTIME} = \text{NP}$ . Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

## L PROOF OF THEOREM 4

**Theorem 4.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O} \left( \frac{1}{1-k_f} + \ln \left( \frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal  $\delta$ -sufficient reason for  $f$ , assuming feature independence. In contrast, unless  $\text{PTIME}=\text{NP}$ , there is no bounded approximation for computing a local cardinally minimal  $\delta$ -sufficient reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

*Proof.* We divide the proof into two lemmas: one establishing the approximation guarantee for the global case, and the other demonstrating the intractability of the local case.

**Lemma 12.** *Given a neural network or tree ensemble  $f$  and an empirical distribution  $\mathcal{D}$  over a fixed dataset  $D$ , Algorithm 2 yields a constant  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal  $\delta$ -sufficient reason for  $f$ , assuming feature independence.*

*Proof.* The proof will follow a similar approach to that of Lemma 10, where we showed that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of  $\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)$ . After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by  $\mathcal{O}(\ln(|D|))$ .

Similar to Lemma 10, we will again prove a stronger claim — namely, that the result holds for *any* model, assuming the trivial condition that its inference time is polynomially computable, and the additional condition concerning the removal of redundant features, as described in Lemma 10. As discussed there, both conditions are satisfied by our neural network and tree ensemble formalizations.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs  $\mathbf{x}, \mathbf{z}$  in the dataset  $D$ , as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

Unlike Lemma 10, where the approximation relied on the result by Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the setting here requires a different condition due to the supermodular nature of the function. Specifically, Shi et al. (Shi et al., 2021) provided an approximation guarantee of  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ , where  $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$ . We established in Proposition 2 that the value function  $v_{\text{suff}}^g$  is monotone non-decreasing, and under the assumption of feature independence, Proposition 3 further shows that it is supermodular. Given that Algorithm 2 runs in polynomial time, this leads directly to an approximation guarantee of  $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ , where  $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$ .

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from  $f$  — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature  $i$  is strictly redundant if and only if  $v(\{i\}) - v(\emptyset) = 0$ . This preprocessing yields a new function  $f'$  that behaves identically to  $f$  over the remaining features and ensures that, for every  $i$ , both  $v_{\text{suff}}^g(\{i\}) > 0$  and  $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$  hold.

In order for us to show that this expression is both bounded and constant, similarly to Lemma 10 we will perform the exact same preprocessing phase (which, as we explained there, can be performed in polynomial time both for neural networks as well as tree ensembles) as before where we remove all redundant features from  $f$ . We note that here too, it holds that a feature  $i$  is strictly redundant if and only if  $v(\{i\}) - v(\emptyset) = 0$ . Hence, this preprocessing phase will give us a new function  $f'$  for which it both holds that it is equivalent to the behaviour of  $f$  for all remaining features and also satisfies that for any  $i$  it holds that both  $v_{\text{suff}}^g(\{i\}) > 0$  and  $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$  hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by  $\frac{1}{|D|}$  and its expected value by  $\frac{1}{|D|^2}$ , we obtain that  $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$  and  $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$ . This implies, as in Lemma 10, that the first term is lower bounded by  $\mathcal{O}(\ln(|D|))$ , and for the second term we derive the following:

Since  $v_{\text{suff}}^g$  is supermodular (Proposition 3), we have that  $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$ . It follows that  $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$ . This implies:  $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$ , and therefore,  $1 - |D|^2 \leq k^f \leq 0$ . This yields  $\frac{1}{1-k^f} \leq |D|^2$ , demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly smaller in practice, our goal here is simply to show that it remains constant — unlike in the local setting, where no bounded approximation is achievable.

□

**Lemma 13.** *Unless  $PTIME=NP$ , there is no bounded approximation for computing a local cardinally minimal  $\delta$ -sufficient reason for any  $\langle f, \mathbf{x} \rangle$ , even when  $|D| = 1$ .*

*Proof.* The proof follows a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local  $\delta$ -sufficient reason for some  $\langle f, \mathbf{x} \rangle$ , where  $f$  is either a neural network or a tree ensemble—even when  $|D| = 1$ . Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline  $\mathbf{z} = 0_n$ , deciding whether a sufficient reason exists is NP-hard unless  $PTIME = NP$ . Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

## M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

**Proposition 1.** *Assume a neural network  $f : \mathbb{F} \rightarrow [c]$  with ReLU activations, some dataset  $\mathbf{D}$ , such that  $\mathcal{D}$  is the empirical distribution defined over  $\mathbf{D}$ , and some threshold  $\delta \in \mathbb{Q}$ . Then unless  $NP \subseteq DTIME(n^{\mathcal{O}(\log(\log(n)))})$ , there does not exist a polynomial-time  $\mathcal{O}(\ln(|\mathbf{D}|) - \epsilon)$  approximation algorithm for computing a cardinally minimal global contrastive reason for  $f$  concerning  $\mathcal{D}$ , for any fixed  $\epsilon > 0$ .*

*Proof.* The proof proceeds via a polynomial-time approximation-preserving reduction from a specific variant of Set Cover that is, similarly to regular Set Cover, provably hard to approximate within the factor given in the proposition (Vazirani, 2001).

### Partial Set Cover:

**Input:** A universe  $U := \{u_1, \dots, u_n\}$ , a collection of sets  $S := \{S_1, \dots, S_m\}$  such that for all  $j$   $S_j \subseteq U$ , and a coverage threshold  $\delta \in \{0, 1, \dots, n\}$ .

**Output:** A minimum-size subset  $C \subseteq S$  such that  $|\cup_{S_j \in C} S_j| \geq \delta$ .

We reduce this problem to that of computing a *global*, cardinally-minimal contrastive reason  $S$  for a neural network  $f$  with respect to a distribution  $\mathcal{D}$  defined over a dataset  $\mathbf{D}$ , and some threshold  $\delta'$ . We begin by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset  $\mathbf{D}$ . The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over  $\mathbf{D}$ , this implies that:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left( \frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) \end{aligned} \quad (74)$$

where we denote  $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$ . We now construct a ReLU neural network  $f$ . Before doing so, we specify the functional form that we intend the constructed neural network  $f$  to implement. For each set  $S_j$ , we introduce two vectors  $\mathbf{x}^j$  and  $\mathbf{y}^j$ , both belonging to  $\mathbb{Z}^n$  and drawn from the dataset  $\mathbf{D}$ , such that they agree on all coordinates outside  $S_j$ ; that is, they share identical feature values on the complement  $\bar{S}_j$ . Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in  $S_j$  by selecting them from two distinct vectors  $\mathbf{z}$  and  $\mathbf{z}'$  such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = \mathbf{y}^j \quad (76)$$

We aim to construct  $f$  such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}^j) \quad (77)$$

Let us now assume that both  $\mathbf{x}^j$  and  $\mathbf{y}^j$  occur *uniquely* in the dataset  $\mathbf{D}$  across all  $[n]$  coordinates — that is, no other vector in  $\mathbf{D}$  matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] = \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left( \frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \right) = \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs —  $(\mathbf{x}^j, \mathbf{z}^j)$  and  $(\mathbf{z}^j, \mathbf{x}^j)$  — contribute to the respective sums.

We repeat this construction for each  $S_i$  in the set-cover instance, each time choosing two *distinct* vectors  $\mathbf{x}^i$  and  $\mathbf{y}^i$  that differ on *every* coordinate in  $[n]$ . In total, this yields  $2m$  vectors that form the dataset  $\mathbf{D}$  from which we construct  $f$ .

We then construct a neural network  $f$  that simulates this function. This is achieved using a known result on *neural network memorization* (Vardi et al., 2022), which guarantees that a ReLU network capable of representing any finite set of input–output pairs can be built in polynomial time. Using this result, we construct  $f$  accordingly, and set  $\delta' := \frac{\delta}{|\mathbf{D}|}$ .

We now argue that a subset  $S_j$  is a global contrastive reason for  $f$  under tolerance  $\delta'$  if and only if  $C$  is a partial set cover for  $U$  under tolerance  $\delta$ . By Equation 78, each constructed pair of vectors  $(\mathbf{x}^j, \mathbf{y}^j)$  contributes exactly  $\frac{1}{|\mathbf{D}|}$  to the value term. Moreover, because each  $S_j$  uses its own distinct pair of vectors in  $\mathbb{Z}^n$ , there is no overlap between these pairs. Consequently, the coverage achieved by any set cover  $C$  corresponds exactly to the accumulated contribution of the selected subsets in the input domain of  $f$ . This completes the reduction.

□

## N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

**Lemma 14.** *If  $f$  is an orthogonal DNFs and the probability term  $v_{\text{suff}}^g$  can be computed in polynomial time given the distribution  $\mathcal{D}$  (which holds for independent distributions, among others), then obtaining a subset-minimal global  $\delta$ -sufficient reason can be obtained in polynomial time.*

*Proof.* We begin by defining orthogonal DNFs (Crama & Hammer, 2011) before presenting the full proof.

**Definition 2** (Orthogonal DNFs). *A Boolean function  $\varphi$  is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms  $T_1 \vee T_2 \vee \dots \vee T_m$ , such that for every pair of distinct terms  $T_i$  and  $T_j$  ( $i \neq j$ ):*

$$T_i \wedge T_j \models \perp$$

*This means that no single variable assignment can satisfy more than one term simultaneously.*

The proof mirrors the argument used in Theorem 1, relying on the fact that  $v_{\text{suff}}^g$  can be computed in polynomial time for any set  $S$ .

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2268 For a given set  $S$ , let  $\mathcal{C}^S T_i$  be the constraints that the term  $T_i$  imposes on the variables in  $S$ .  
 2269 We enumerate all pairs of terms  $(T_i, T_j)$ . Any pair whose constraints on  $S$  are *inconsistent* (i.e.,  
 2270  $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$ ) is ignored, since it contributes zero to  $v^g_{\text{suff}}$ .  
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2272 For each *consistent* pair of terms  $(T_i, T_j)$ , we evaluate  $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$ , where  $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$ .  
 2273 The contribution of this pair to  $v^g_{\text{suff}}$  is then  
 2274

$$2275 \Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

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 2277 Under feature independence, these probabilities factor as  $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$ . Since there are at  
 2278 most  $m^2$  such term pairs, the value of  $v^g_{\text{suff}}$  can be computed in polynomial time under an independent  
 2279 distribution  $\mathcal{D}$ .  $\square$   
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## 2281 O DISCLOSURE: USAGE OF LLMs

2282  
 2283 An LLM was used exclusively as a writing assistant to refine grammar and typos and improve clarity.  
 2284 It did not contribute to the generation of research ideas, study design, data analysis, or interpretation  
 2285 of results, all of which were carried out solely by the authors.  
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